

## ALTIRIG · DESIGN DECISION RECORD

# AltiRig TRL 2 review decisions

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## DOCUMENT CONTROL

|             |                                                            |         |                |
|-------------|------------------------------------------------------------|---------|----------------|
| Document ID | ALR-DDR-001                                                | Version | 0.1            |
| Status      | Draft                                                      | Date    | 2026-10-03     |
| Author      | Amish Chadha                                               | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                                     |         |                |
| Repository  | github.com/BoujeeEnjinia1701/altirig                       |         |                |
| Source      | docs/decisions/0001-trl2-review-decisions.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                           |
|---------|------------|--------------|----------------------------------------------------------------------------------|
| 0.1     | 2026-10-03 | Amish Chadha | TRL 2 review decisions D1 to D14, decided under Amish's 2026-10-03 pre-approvals |

## 0001: TRL 2 review decisions

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- **Date:** 2026-10-03
- **Status:** decided. Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." And for this second batch, Amish, 2026-10-03: "Proceed with the remaining 15 scaffolds". Each decision below is the recommendation made at the TRL 2 review and is decided as recommended. Requirements that are not met or at risk are not decided here; they are posed to Amish in `docs/REVIEW.md` and the design decisions register.

### ■ Context

The scaffold (ALR-PRC-001 v0.1) described a sealed vessel with a thrust stand inside and left open the vessel type, whether to chill the air, how to deal with the propeller's recirculating wake, and whether a propeller and motor kit belongs in this repo. The TRL 2 review settled these so the TRL 3 calculations (ALR-CAL-001), the model and the build plan could proceed. Choices that touch safety take the conservative option, and each says what evidence would relax it.

**Safety:** AltiRig is a vacuum vessel with a spinning propeller inside, run from mains power. These decisions set the design basis on paper only. Nothing here makes the chamber safe to use; it is not certified pressure equipment or test equipment.

### ■ Decisions

*Table 1. TRL 2 review decisions.*

| #   | Decision                                                                                                                                                                                                                   | Reason                                                                                                                                                                                    | Affects                               |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| D1  | Match air density, not pressure: R1's 0.74 kg/m <sup>3</sup> is reached at about 61.9 kPa at 20 °C; pressure and temperature are logged with every point                                                                   | Thrust and power scale with density; at room temperature, 54 kPa would give the density of about 6,100 m                                                                                  | R1; controller; ALR-CAL-001 A         |
| D2  | Vessel: a horizontal 813 mm (32 in) x 6.35 mm steel pipe offcut, 1,500 mm long, with two bought 2:1 ellipsoidal tank heads; one head welded on the rear, one on a flange ring as the door                                  | A thin steel drum collapses at a few kPa of vacuum; a flat-panelled box needs heavy ribs; pipe and tank heads are stocked items a welding shop can join                                   | Vessel parts; R5, R7; cost            |
| D3  | Design the vessel for full vacuum (101.3 kPa difference) and apply R7's factor of 4 at full vacuum, not only at the working point                                                                                          | Conservative: the pump can reach near full vacuum if the relief valve sticks. Relax only with a code calculation by a qualified engineer and a strain-gauged proof test                   | R7; ALR-CAL-001 B                     |
| D4  | Add R12: an adjustable vacuum relief valve opens at 55 kPa below atmosphere, so the chamber cannot go below 46.3 kPa absolute in normal use                                                                                | Covers every working point (lowest 57.4 kPa) with margin and limits the load on the window and the door                                                                                   | R12; valve manifold                   |
| D5  | Add R11: the motor can be armed only with the door latched (interlock switch) and the key turned; the emergency stop removes motor and pump power; the bleed valve is normally open, so the chamber vents on loss of power | The door is held shut by air under vacuum, so the hazard is a running propeller when the door is open or a run that cannot be stopped                                                     | R11; cabinet; build plan safety stops |
| D6  | Window: clear polycarbonate 20 mm, never acrylic or glass, placed with its nearest edge more than 15° from the propeller plane                                                                                             | Polycarbonate yields rather than shatters; keeping it out of the fragment zone protects the one non-steel wall. Relax only with impact tests of the window at TRL 4                       | Window; R7, R11                       |
| D7  | Thrust stand: two spring-steel flexure leaves (no rails), a 10 kg bending-beam cell for thrust, a torque head on two bearings with an arm on a 5 kg cell, an optical speed sensor                                          | No sliding friction; bought C3 cells; the share carried by the leaves is fixed and calibrated out                                                                                         | R3, R4; stand parts                   |
| D8  | Room temperature only in the first version; chilling towards -20 °C is a later variant                                                                                                                                     | Density, not temperature, sets thrust; chilling a 0.9 m <sup>3</sup> vessel adds a refrigeration plant and condensation                                                                   | Scope; R1                             |
| D9  | Closed chamber with a removable 63 % open perforated baffle, and side-by-side comparison runs at ambient pressure published with every data set                                                                            | The scaffold's open question on recirculation; the baffle breaks up the jet and swirl, and the comparison runs measure what the loop does. Whether R6 is met is posed to Amish separately | R6; baffle; ALR-CAL-001 F             |
| D10 | No lithium or sodium-ion battery inside the chamber in the first version; the motor is powered from a 1,500 W supply outside through the feed-through                                                                      | Conservative: a battery fire in a closed vessel raises the pressure and fills it with toxic fumes. Relax only with a vent, a fire-containment plan and abuse tests of the pack type       | Scope; ColdCell; feed-through         |
| D11 | Co-design: the first candidate partner to approach is the Department of Aerospace Engineering, IIT Kanpur; the second is the Department of Mechanical Engineering,                                                         |                                                                                                                                                                                           | ALR-PRB-001 co-design                 |

| #   | Decision                                                                                                                                                                                                                              | Reason                                                                                                              | Affects                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|     | Kathmandu University. First candidate region: the Nepal and Indian Himalaya. Recorded as candidates, not agreed                                                                                                                       | Propeller facilities and Himalayan users; a partner reviews the vessel calculation and runs the open-air comparison |                                                    |
| D12 | The high-altitude propeller, motor and tuning kit is not part of this repo; AltiRig publishes measured data and the Kitewright Lift and Range frames publish their kits from it                                                       | Keeps AltiRig a test rig and the kits with the aircraft they serve                                                  | Scope; shared blocks                               |
| D13 | Shared blocks: CalRig's proof-load method calibrates the load cells; ColdCell packs are not tested inside the chamber in this version (D10)                                                                                           | Re-use across the portfolio without adding a battery hazard                                                         | ALR-PRC-001 shared blocks                          |
| D14 | Publish as an open engineering reference with a notice that it is not certified pressure or test equipment; before any use a qualified engineer reviews the vessel calculation and the first pump-down is a proof test behind a guard | Conservative; follows the safety notes in the scaffold                                                              | README and every document; build plan safety stops |

## ■ Consequences

- R11 and R12 are added to ALR-REQ-001.
- The vessel, welding, motor supply and safety controls take the estimated cost well above the USD 1,000 value-engineering target (ALR-DEC-001, Value engineering). Amish accepted cost overruns in the same instruction; R10 is posed to Amish with options in `docs/REVIEW.md`.